

IMPACT STATEMENT CARD: AGRICULTURAL EXTENSION RETRIEVAL DATASET

Project Title: Optimising RAG Document Retrieval for Agronomic Advice

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Purpose of the Project

Our project builds the retrieval engine behind a Retrieval-Augmented Generation (RAG) system, designed to connect smallholder farmers' agricultural questions with the most relevant expert advice from a knowledge base of 695 factsheets. The core community problem is that reliable agronomic knowledge already exists, covering crop diseases, pests, nutrient deficiencies, and climate adaptation, but is scattered across documents that are hard to search in time, especially when a farmer needs an answer quickly to protect a threatened crop.

Positive Impacts

A working retrieval system could meaningfully reduce the time it takes for a farmer, or the extension worker supporting them, to find the right advice for a specific problem. Given that Nigeria is the most represented country in our dataset (133 of 695 documents), alongside Ghana, Tanzania, and Kenya, communities in these better-covered regions stand to benefit most directly, since faster, more accurate information can mean catching an issue before it costs a household its harvest.

Potential Risks and Harms

The clearest risk is retrieval error, particularly near miss documents that describe a similar but incorrect issue, which could lead someone toward the wrong treatment if trusted uncritically. There is also a fairness risk tied to uneven country representation, countries with only a handful of documents, such as Togo (3) or Benin (4), would likely receive noticeably weaker retrieval performance, meaning the tool could unintentionally serve already better-documented regions more effectively than others. Economically, acting on a wrong recommendation carries real cost, wasted money on the wrong treatment or a damaged crop that could have been saved.

Values

This project is guided by accuracy over convenience, accessibility of existing knowledge, caution around unintended harm, and groundedness in real, verifiable sources. We protect farmer autonomy and dignity by deliberately scoping our system to retrieving supporting documents rather than generating final advice, ensuring human judgement remains the final decision-maker rather than being replaced by an AI-generated answer.

Mitigation Strategies

We are testing multiple retrieval methods, including keyword-based and meaning-based approaches, specifically to catch near miss documents that a single method might miss. We plan

to review queries where our system performs poorly to check whether weaker performance clusters around specific countries or crops, documenting these gaps rather than hiding them. We are also being transparent about our dataset's composition, including its heavy reliance on synthetic content, so users of our work understand its current boundaries.

Limitations and Uncertainties

A majority of our documents (637 of 695) are synthetically generated rather than drawn from verified real-world sources, so we cannot fully vouch for their factual accuracy. We also do not know whether our relevance labels were created by agricultural domain experts, since this isn't disclosed in the competition data. Strong performance on this fixed dataset does not guarantee similar reliability on real farmer questions outside of it. This project should be understood as a proof of concept for retrieval quality, not a system ready for real-world deployment without further expert review and farmer testing.